



**COLLEGE OF PURE AND APPLIED SCIENCES
COMPUTER SCIENCE DEPARTMENT**

Session: 2024/2025

Credit Unit: 3

Course Title: Computer System Performance Evaluation

Course Code: CSC 415

Time: 3 Hours

Instructions

- Answer any 4 out of the 6 questions below.
- Each question carries 15 marks (divided into three sub-parts, 5 marks each).
- Focus on applied problem-solving and real-world scenarios while demonstrating both theoretical knowledge and practical relevance.

QUESTION ONE (15 MARKS)

An online retailer needs to check how well its web server handles busy shopping periods.

- (a) Propose a simple plan to measure the server's response time and throughput using available tools. (5 marks)
- (b) Identify two issues that might affect these measurements (for example, network delays or traffic spikes) and suggest one fix for each. (5 marks)
- (c) Choose two key indicators (like average response time and requests per second) to monitor the server's performance and explain how to collect these metrics. (5 marks)

QUESTION TWO (15 MARKS)

A company wants to test a new database system under different workloads.

- (a) Propose an approach to benchmark the database using both simulated (synthetic) and real workload data. (5 marks)
- (b) Describe a simple way to simulate a workload that the database might see in production. (5 marks)
- (c) Select two performance metrics (for example, query response time and transaction rate) and explain how to measure them. (5 marks)

QUESTION THREE (15 MARKS)

A router in a network sometimes causes delays in packet delivery. Assume the router can be modeled as an M/M/1 queue, where the utilization (ρ) and the average delay per packet (W) are given by:

$$\rho = \frac{\lambda}{\mu}$$
$$W = \frac{1}{\mu - \lambda}$$

Here, λ represents the packet arrival rate (packets per second) and μ represents the packet service rate (packets per second).

- (a) Use the above equations to estimate the router's utilization and average delay when $\lambda=50$ packets/sec and $\mu=100$ packets/sec. (5 marks)
- (b) Explain what happens to the average delay if the rate of incoming packets increases, keeping μ constant. Provide a numerical example. (5 marks)
- (c) Suggest one simple method to reduce the delay in packet processing. (5 marks)

QUESTION FOUR (15 MARKS)

A data center manager wants to test server performance during high load.

- (a) Outline a simple simulation experiment (using a tool or manual calculations) to study server response under heavy load. (5 marks)
- (b) Identify two key parameters (like job arrival rate and processing time) to vary during the simulation and explain why. (5 marks)
- (c) Propose a way to check if the simulation results match real performance data. (5 marks)

QUESTION FIVE (15 MARKS)

A cloud service provider is looking to improve its overall system speed.

- (a) Propose a simple performance model that shows how different components (CPU, disk, network) affect system throughput. (5 marks)
- (b) Identify one main bottleneck (for example, slow disk I/O) and explain its effect on performance. (5 marks)
- (c) Suggest one easy solution to reduce the impact of this bottleneck. (5 marks)

QUESTION SIX (15 MARKS)

A company is evaluating a distributed system that spans several servers.

- (a) Propose a simple method to measure communication delay and data transfer rates between servers. (5 marks)
- (b) Explain how you would check if work is evenly distributed across all servers. (5 marks)
- (c) Select two overall performance metrics (such as total throughput and average response time) and explain how to gather these metrics. (5 marks)

process



Department of Computer Science
CSC 433: Human Computer Interactions
2024/2025 First Semester Examination

Time: 3 hrs.

3 units

Marks: 60

INSTRUCTION: ANSWER ANY FOUR QUESTIONS.

QUESTION ONE

- I. Briefly explain Human Computer Interface (HCI) 5 marks
- II. HCI is a multidisciplinary subject covering computer science, behavioral sciences, cognitive science, ergonomics and human factors, psychology, information security, speech language pathology, ethnography, engineering and design principles. Write a short note on how HCI interfaces with Behavioral sciences. 5 marks
- III. Justify the reason for designing human-computer interfaces 5 marks

QUESTION TWO

- I. Explain what is meant by "innovative interfaces in everyday life". 5 marks
- II. Discuss data display guidelines. 5 marks
- III. Briefly explain human hearing as it relates to computing 5 marks

QUESTION THREE

- I. Explain the steps involved in screen design. 5 marks
- II. One of the most important effects which has to be achieved when presenting information is to draw users' attention to important items. Discuss the techniques or visual attributes that can be used to highlight data. 5 marks
- III. Explain human learning and memory as it relates to computing. 5 marks

QUESTION FOUR

- I. Sitting too long is associated with a higher risk of being diagnosed with breast, colon, colorectal, endometrial and ovarian cancers. Explain a way out of this in relation to human Computer Interactions. 7 marks
- II. How does posture impact positivity? 4 marks
- III. What is command line interface? 4 marks

QUESTION FIVE

- I. Data entry is notoriously error prone; hence, data validation attempts to check that all mandatory fields are filled in and that the data entered is correct, or at least reasonable. Explain three of the commonly used methods for data validation. 6 marks
- II. Describe any three alternative data entry techniques you are familiar with. 6 marks
- III. Briefly explain graphical displays. 3 marks

QUESTION SIX

- I. What is interface structure design? 3marks
- II. How can user experience be improved through interface design? 5 marks
- III. Explain task analysis. 7 marks

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COLLEGE OF PURE AND APPLIED SCIENCES (COPAS)
DEPARTMENT OF COMPUTER SCIENCE

SESSION: 2024/2025

COURSE CODE: CSC 411

COURSE TITLE: Organization of Programming Languages

INSTRUCTION: Answer FOUR questions only

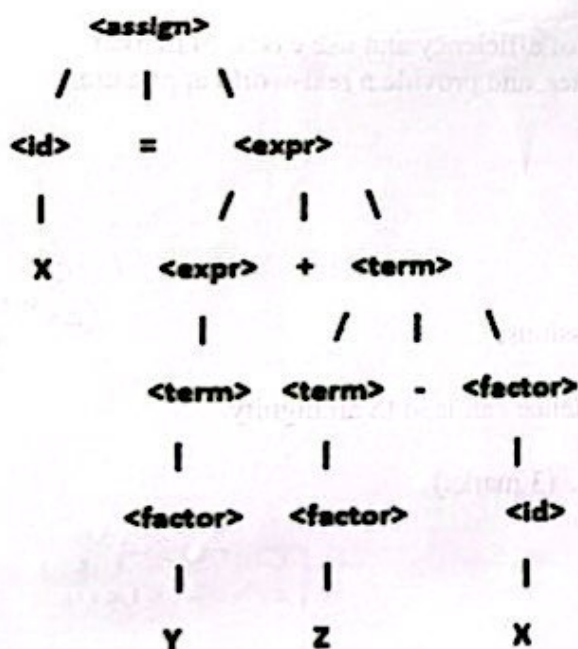
SEMESTER: First
CREDIT UNIT: 3

TIME: 2 hours

QUESTION 1

- Write a short note on the following: (6 marks)
 - imperative paradigm
 - functional paradigm
 - logical paradigm
 - object-oriented paradigm
- What do you understand by inheritance and polymorphism as associated with object-oriented programming? (3 marks)
- Explain the following structural layers of programming languages: (6 marks)
 - lexical
 - syntactic
 - contextual
 - semantic.

QUESTION 2



assign ::=
 expr ::=
 term ::=
 factor ::=
 id ::= X Y Z

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- a. Write out the grammar used in generating the parse tree given above. (6 marks)
- b. What is Recursive-Descent Parser? (3 marks)
- c. Describe the complexity of parsing algorithms. (3 marks)
- d. State the two classes of parsers with their function. (3 marks)

state transition

lexer

QUESTION 3

- a. What is a lexical analyzer? (3 marks)
- b. Explain the three main approaches to building a lexical analyzer. (6 marks)
- c. What are the two distinct goals of syntax analysis? (2 marks)
- d. Compare top-down parsing and bottom-up parsing with examples. (4 marks)

lexer

12
ADP

QUESTION 4

- a. What is an interpretative language and how does it differ from a compiled language? (4 marks)
- b. Discuss two advantages and disadvantages of using an interpretative language. (4 marks)
- c. Give two examples of an interpretative languages and describe their primary use cases. (4 marks)
- d. What role does bytecode play in languages like Java? (3 marks)

memory location
12
12

QUESTION 5

- a. What is the difference between primitive and non-primitive data types? Provide examples. (3 marks)
- b. Explain how dynamic and static typing affect memory management in programming languages. (4 marks)
- c. Compare and contrast arrays and linked lists in terms of efficiency and use cases. (4 marks)
- d. Describe the key differences between stacks and queues, and provide a real-world application for each. (4 marks)

QUESTION 6

- a. What is an ambiguous grammar? (3 marks)
- b. Consider the following grammar for arithmetic expressions:
 $E \rightarrow E + E \mid E * E \mid id$
With an example parse tree, show how the below sentence can lead to ambiguity.
 $id + id * id$ (9 marks)
- c. Explain how ambiguity in a grammar can be resolved. (3 marks)

lookup table

FIFO - queue
LIFO - stack

using regular expressions and automata



**COLLEGE OF PURE AND APPLIED SCIENCES
COMPUTER SCIENCE DEPARTMENT**

Session: 2024/2025

Credit Unit: 3

Course Title: Artificial Intelligence

Course Code: CSC 405

Time: 3 Hours

Instructions

- Answer any 4 out of the 6 questions below.
- Each question carries 15 marks (divided into three sub-parts, 5 marks each).
- Focus on applied problem-solving and real-world scenarios while demonstrating both theoretical knowledge and practical relevance.

| QUESTION ONE (15 MARKS)

A busy warehouse uses a system that sorts packages automatically with cameras and sensors.

- Propose a plan for an AI system that sorts packages in real time.(5 marks)
- Identify two possible issues that could arise (for example, errors or worker impacts) and suggest a fix for each.(5 marks)
- Choose two measurable outcomes (such as items per minute or error percentage) to check if the system is successful (5 marks)

QUESTION TWO (15 MARKS)

In a multi-level warehouse, robots need to find the shortest path between storage areas.

- Compare how BFS and DFS work in this setting by giving one simple example for each method. (5 marks)
- Show how A* search can improve pathfinding using a distance estimate like Manhattan distance. (5 marks)
- Write a short pseudo-code for BFS on a grid that represents the warehouse layout.(5 marks)

QUESTION THREE (15 MARKS)

A small manufacturing plant uses a logic system to manage machine operations.

- Write a rule that stops a machine from starting if it is already under maintenance.(5 marks)
- Contrast a simple true/false statement (for one machine) with a version that uses a variable to cover multiple machines. (5 marks)
- Apply a basic logical rule (such as modus ponens) to decide to stop a machine when it shows an overheating condition.(5 marks)

| QUESTION FOUR (15 MARKS)

A health service sorts patients into high-risk or low-risk groups using test results and symptoms.

- Lay out a plan that uses patient data (like blood tests and symptoms) to classify risk, and choose a simple model (e.g., Decision Tree).(5 marks)
- Outline the steps to collect data, clean it, build the model, and test it.(5 marks)
- Pick two measures (for example, recall and precision) to check model performance and explain why each is useful in this context.(5 marks)

QUESTION FIVE (15 MARKS)

A smart building uses sensors (motion and temperature) to decide when to turn on the air conditioning.

- (a) Draw or outline a simple Bayesian network with nodes for a motion sensor, a temperature sensor, and room occupancy. (5 marks)
- (b) Use a scenario where one sensor gives uncertain data to decide if the room is occupied (5 marks)
- (c) List two common sensor issues (like drift or failure) and suggest a quick fix for each. (5 marks)

QUESTION SIX (15 MARKS)

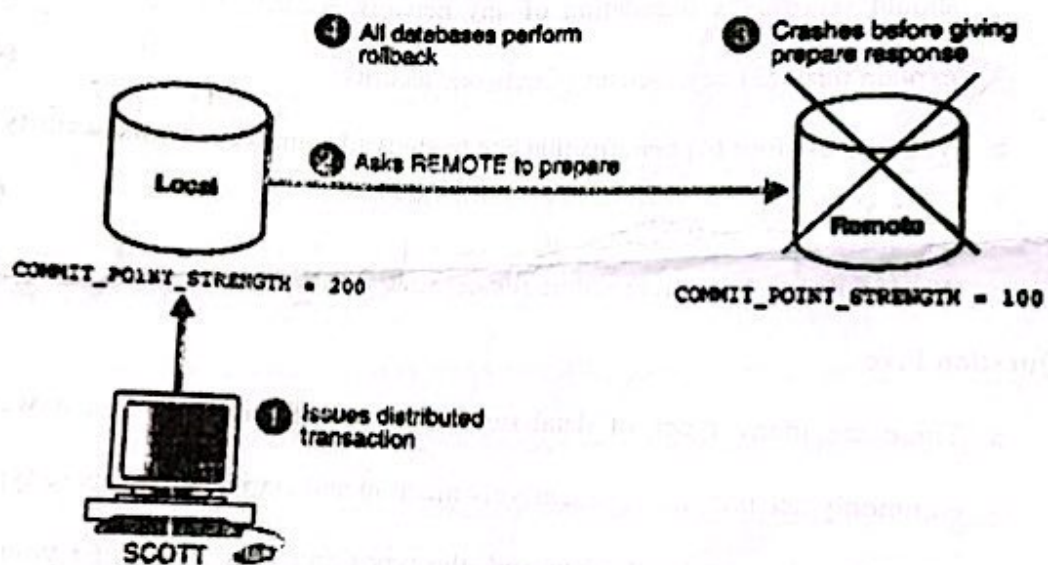
A warehouse cart learns how to deliver items and avoid obstacles by trial and error.

- (a) Set up the RL scenario by listing key states (e.g., location, battery level), actions (e.g., move forward, turn), and rewards (e.g., successful delivery, collision penalty). (5 marks)
- (b) Show how the cart can update its decisions after a good or bad outcome, using a simple RL method like Q-learning. (5 marks)
- (c) Identify two challenges (for instance, changes in layout or limited sensor view) and offer one quick fix for each. (5 marks)

Question One

- a. Bellow figures illustrates the sequence of events when there is a failure during the prepare phase of a distributed transaction. Elucidate the sequence of events when there is a failure during the commit phase and failure during prepare phase of a distributed transaction.

(10 marks)



- b. A distributed transaction includes one or more statements that, individually or as a group. Exhaustively and describe five (5) that can execute DML and DDL statements in parallel, and INSERT direct load statements serially restrictions.

(5 marks)

Question Two

- a. Succinctly describe what you understand by the terms a Parallel Systems. Differentiate between the parallel computing and parallel processing

(5 marks)

- b. Based on your understanding, Flynn's has classified the computer systems based on parallelism in the instructions and in the data streams. Distinguish between the two Single Instruction, Multiple Data stream (SIMD) schemes. Support your answer with diagrams, and explain in details (10 marks)

Question Three

- a. Classify the types of interactions that can be represented in a model. (3 marks)
- b. Explicate the five (5) types of UML diagrams for system modeling. (5 marks)
- c. Draw and describe with example in details data-driven models using UML sequence diagrams? (7 marks)

Question Four

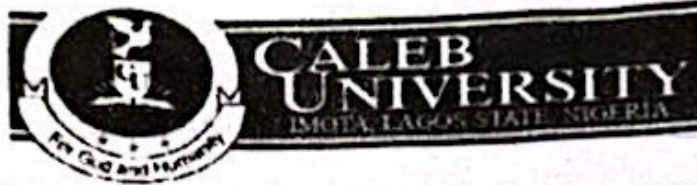
- a. How would you describe the terms network security? There are three key focuses that should serve as a foundation of any network security strategy, identify and concisely explain three (3) key focuses of network security (4 marks)
- b. What are the four (4) benefits that can be derived from the network security? Discuss (4 marks)
- c. What tools can be used to secure the network? Discuss (7 mark)

Question Five

- a. There are many types of database for many different purposes; a web app will most commonly use one. Comprehensively mention and clarify the database types. (5 marks)
- b. What do we mean by the backend, the types and what determine your backend choice? Explain. (6 marks)
- c. Deliberate how cybercrime can be mitigated (4 mark)

Question Six

- a. Enumerate the prerequisite for building a web application (3 marks)
- b. Discuss Wireless in Local Loop (WLL). (3 marks)
- c. Succinctly describe what is the difference of firewalls at Application security and internet security (9 marks)



**COLLEGE OF PURE AND APPLIED SCIENCES (COPAS)
DEPARTMENT OF COMPUTER SCIENCE**

First Semester 2024/2025 Academic Session
Course Title: Introduction to Software Engineering
Course Unit: 4 Units
Instruction: Attempt any four (4) questions.

Course Code: CSC 407/403
Time: 3 hours

Question #1:

- (a) Define Software Engineering and explain its role in ensuring software quality, reducing costs, and managing complexity. (3 marks)
- (b) Discuss the principles of abstraction, decomposition, and modularity in Software Engineering. Provide examples. (4 marks)
- (c) Explain how Software Engineering impacts real-world industries, such as healthcare and finance. (4 marks)
- (d) Discuss the challenges in Software Engineering and suggest strategies to overcome them. (4 marks)

Question #2:

- (a) Explain the concept of the Software Development Life Cycle (SDLC) and its importance. (3 marks)
- (b) Compare and contrast the Waterfall, Agile, and Spiral models of software development. (4 marks)
- (c) Discuss the key factors influencing the selection of a suitable software development model. (4 marks)
- (d) Explain how DevOps integrates development and operations for continuous delivery. (4 marks)

Question #3:

- (a) Differentiate between functional and non-functional requirements with examples. (3 marks)
- (b) Explain three techniques for gathering software requirements and their advantages. (3 marks)
- (c) Describe the components of a Software Requirements Specification (SRS) document. (3 marks)
- (d) Discuss the importance of UML diagrams in requirements engineering. (3 marks)
- (e) Discuss three key legal issues in software engineering and their implications. (3 marks)

Question #4:

- (a) Explain the importance of software architecture in system design. (3 marks)
- (b) Compare Client-Server, Microservices, and Layered Architecture models, highlighting their advantages and disadvantages. (3 marks)
- (c) Define and describe the role of design patterns in software development. Provide examples. (3 marks)
- (d) Discuss the factors influencing the choice of software architecture. (3 marks)
- (e) Explain the importance of intellectual property rights in software development. (3 marks)

Question #5:

- (a) Define software validation and verification, and explain their differences. (3 marks)
- (b) Compare black-box and white-box testing techniques with examples. (3 marks)
- (c) Explain the significance of automated testing and Continuous Integration (CI) in modern software development. (3 marks)
- (d) Discuss the role of Test-Driven Development (TDD) in improving software quality. (3 marks)
- (e) Compare open-source and proprietary software licensing models. (3 marks)

Question #6:

- (a) Define Software Project Management and explain its importance. (3 marks)
- (b) Discuss three key project scheduling techniques used in software development. (3 marks)
- (c) Explain the role of risk management in software projects and the different types of risks. (3 marks)
- (d) Describe the role of Software Configuration Management (SCM) and its benefits. (3 marks)
- (e) Describe how data privacy regulations affect software engineering practices. (3 marks)



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COLLEGE OF PURE AND APPLIED SCIENCES (COPAS)
DEPARTMENT OF COMPUTER SCIENCE

SESSION: 2024/2025

COURSE CODE: CSC 413

COURSE TITLE: COMPUTER GRAPHICS AND VISUALIZATION

INSTRUCTION: ATTEMPT FOUR QUESTIONS IN ALL

SEMESTER: SECOND
CREDIT UNIT: 3

TIME: 3 HOURS

Question One

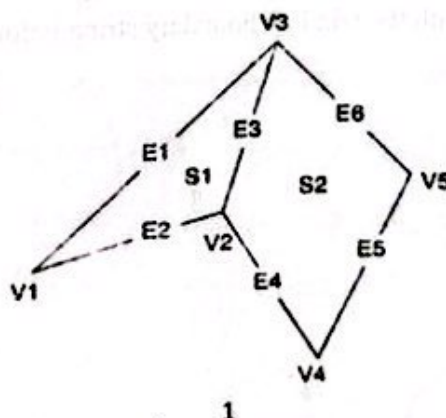
- Describe the steps to be followed in Digital Differential Analyzer scan conversion algorithm? (5 marks)
- By analogy ways through which object can be display on the screen used for raster graphics, differentiate between Raster Scan and Random Scan based on the following factors: (i) Electron Beam (ii) Resolution (iii) Picture Definition (iv) Realistic Display (v) Draw an image (10 marks)

Question Two

- Draw and display bresenham graph from any position (2, 3) (X_k, Y_k) you need to choose between (3, 3) ($X_k + 1, Y_k$) and (3, 4) ($X_k + 1, Y_k + 1$). You would like the point that is closer to the original line. (5 marks)
- Elucidate in detail about the DDA scan conversion algorithm in Computer Graphics? (10 marks)

/ Question Three

- Highlight and describe two (2) 3D object representation. Study this figure below critically and provide the following information table: (i) Vertex table (ii) Vertex table (iii) Polygon surface table (8 marks)



- b. Identify and explain four (4) possible relationships that a surface can have with a specified area boundary. (4 marks)
- c. Explain the following types of computer graphics related to methods in surface detection: (i) Scan Line Method (ii) Area-Subdivision Method (3 marks)

Question Four

- a. Transformations are a fundamental part of computer graphics used to position objects, to shape objects, to change viewing positions. Enumerate and describe two (2) main types of transformation that one can perform in 2 dimensions with the matrix / vector representation of translations with the equation formula (8 marks)
- b. Incline and clarify two (2) major types of parallel projection, display it and sub types (7 marks)

Question Five

- a. Succinctly describe the terms perspective projection in details, and identify three (3) types of perspective projection with the aid of flow diagram projection plane. (4 marks)
- b. Itemize four (4) properties of bezier curve (4 marks)
- c. Enumerate five (5) properties of b-spline curve (5 marks)
- d. The Bezier curve can be represented mathematically. Express the meaning of the following mathematical sentences: (i) $f(x, y) = 0$ (ii) $x^2 + y^2 - R^2 = 0$ (2 marks)

Question Six

- a. Draw and clarify in details point clipping rectangle window, indicates X and Y-coordinate lies in between W_x1 and W_y1 . (5 marks)
- b. Describe the steps in Cyrus-Beck Line Clipping Algorithm. (7 marks)
- c. Various techniques are used to provide text clipping in a computer graphics, identify three (3) methods for text clipping with the window boundary string before and after (3 marks)



COLLEGE OF PURE AND APPLIED SCIENCES (COPAS)
DEPARTMENT OF COMPUTER SCIENCE

SESSION: 2023/2024

COURSE CODE: CSC 439

COURSE TITLE: NET-CENTRIC COMPUTER

INSTRUCTION: ATTEMPT FOUR QUESTIONS IN ALL

SEMESTER: FIRST
CREDIT UNIT: 3

TIME: 2 ½ HOURS

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Question One

- Elucidate what you understand by the term mobile and wireless computing, support your answer with example explain in details ✓ (3 marks)
- Exhaustively and describe five (5) advantages of mobile computing ✓ (5 marks)
- Contribute a brief explanation of Near Field Communication as one of the current trends in mobile computing ✓ (3 marks)
- Why is the Mobile Software important in Mobile and Cloud Computing? (4 marks)

Question Two

- Succinctly describe what you understand by the terms a distributed system, and clarify two (2) general ways that distributed systems function ✓ (3 marks)
- Based on your understanding, provide the advantage of distributed computing, and designate two (2) service orientations ✓ (4 marks)
- Which of the security infrastructure is most critical and why? (4 marks)
- Briefly illuminate in details the following types: (i) Web 2.0 technologies (ii) Virtualization ✓ (4 marks)

Question Three

- Deliberate explain four (4) advantages of client Server computing (4 marks)
- In client-server computing, the client requests a resource and the server provides that resource. Explicate the five (5) characteristics of client server computing (5 marks)
- What makes the Client Server configuration peculiar from the Peer-to-peer? (6 mark)

Question Four ✓

- a. How would you describe the terms network security? There are three key focuses that should serve as a foundation of any network security strategy, identify and concisely explain three (3) key focuses of network security ✓ (4 marks)
- b. What are the four (4) benefits that can be derived from the network security? Discuss (4 marks)
- c. What tools can be used to secure the network? Discuss ✓ (7 mark)

Question Five

- a. There are many types of database for many different purposes; a web app will most commonly use one. Comprehensively mention and clarify the database types. (5 marks)
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CALEB UNIVERSITY
COLLEGE OF PURE AND APPLIED SCIENCES (COPAS) DEPARTMENT OF
COMPUTER SCIENCE

2023/2024 Session First Semester Examinations

CSC 415: Computer Performance and Evaluation

Time Allowed: 2 hours

Instruction: Answer any four (4) questions

1(a) State the function of performance evaluation in modern computer systems? (2marks)

(b) For a set of processes with arrival times, burst times, and priorities as follows:

Process 1: Arrival Time = 0, Burst Time = 3, Priority = 2

Process 2: Arrival Time = 1, Burst Time = 5, Priority = 1

Process 3: Arrival Time = 2, Burst Time = 4, Priority = 3

Using priority scheduling, calculate the average waiting time. (4marks)

(c) A processor executes 4000 instructions in 2 milliseconds. What is its throughput in instructions per second?

(3marks)

(d) (i) Mention and describe a paradigm you are familiar with - for programming and execution. (2marks)

(ii) State three (3) impact of upgrading RAM computer performance. (3marks)

(e) Describe the smallest unit of a computer program with respect to its function. (1mark)

2(a) (i) State Amdahl's Law. (1mark)

(ii) State Moore's Law. (1mark)

(b) Describe the approach that is often being utilized to increase a computer system's central processor unit's throughput. (2marks)

(c) If a system processes 500 tasks in 5 minutes, what is its throughput? (3marks)

(d) (i) If a web server takes 150 milliseconds to respond to a request and the network latency is 50 milliseconds, what is the total response time? (2marks)

(ii) Identify the primary determinant of the tools utilized in computer performance assessment. (1mark)

(iii) Highlight three (3) main aspects are usually examined during computer performance evaluation. (3marks)

(e) State any two (2) steps that can be taken when a computer takes a long time to boot up. (2marks)

3(a) In terms of function describe two (2):

(i) Network performance tools. (2marks)

(ii) Application performance monitoring tools. (2marks)

(iii) Load testing tools. (2marks)

(iv) Profiling Tools.

(b) What is the function of the Kubernetes Dashboard?

(c) A database query takes 250 milliseconds to execute on the server. If the client-side processing time is 30 milliseconds, what is the total response time?

(d) (i) In terms of its function, describe the technique known as "auto-scaling"?

(e) If a program executes 1000 instructions with an average of 2 clock cycles per instruction (CPI), what is the total execution time?

4(a) Describe two (2) activities involved in using the Benchmarking Software.

(b) (i) Highlight three (3) tools used in "Disk Benchmarking"

(ii) You have been hired by an organization as a technologist and as instructed to carry out memory performance measurement on all the systems in the organization (consisting of Windows operating system-based and Mac operating system-based laptops). Briefly describe the two major tasks you will perform in using the Task Manager/Activity Monitor as a tool in achieving your purpose.

(c) Providing not less than two points for each, describe the following general guidelines on how to interpret and explain results produced by performance measurement.

(i) Define the Context (ii) Identify Key Metrics (iii) Presenting Results

(d) State the reason why "Thermal Throttling" occurs in a computer system.

5(a) Describe the experiments that can be carried out to measure each of the following aspects of computer performance.

(i) Memory Performance

(ii) Virtualization Performance

(b) (i) If the CPI of a program is reduced from 4 to 2 and the number of instructions remains the same (1000), what is the reduction in execution time?

(ii) List and explain three (3) benefits of overclocking.

(c) Explain the following methods of performing overclocking.

(i) Cooling Solutions

(ii) Gradual Approach

(d) State (2) risks/considerations of overclocking.

6(a) Explain the concept of preventive maintenance in the context of computer systems.

(b) What role does proper cable management play in computer maintenance?

(c)(i) A web server handles 5000 requests in 10 minutes. What is its throughput in requests per second?

(ii) Define the term 'defragmentation'

(iii) What is the significance of defragmentation as related to computer performance?

(iv) State the importance of carrying out regularly update software and operating systems

(d) How does a registry cleaner contribute to improving the performance of a computer?



CALEB UNIVERSITY
COLLEGE OF PURE AND APPLIED SCIENCES (COPAS)
DEPARTMENT OF COMPUTER SCIENCE
2023/2024 Session First Semester Examinations

CSC 405 Artificial Intelligence

Instruction: Answer any four (4) questions

Time Allowed: 1 hour 45mins

- ✓ 1(a) (i) What do you understand by the term Artificial Intelligence? (1mark) ✓
- (ii) In recent times most human activities are now being performed by A.I. Highlight (4) tasks typically performed by humans that is now being generally performed by Artificial Intelligence? (4marks) ✓
- (b) Artificial Intelligence can be classified into different categories. Describe two (2) broad categorizations of Artificial Intelligence. (2marks) ✓
- (c) Give a short description of each of the following sub-fields of Artificial Intelligence. (3marks)
- (i) Machine Learning (ML) ✓ (ii) Natural Language Processing (NLP) ✓ (iii) Computer Vision ✓
- (d) (i) Highlight ~~five~~ (4) industries where Artificial Intelligence has its applications. (4marks) ✓
- (ii) State the goal of knowledge representation (KR). (1mark) ✓
- 2(a) What kind of system can be referred to as an "Intelligent Agent"? ✓ (1mark)
- (b) (i) Give an explanation of the four (4) essential elements and traits of intelligent agents. (4marks) ✓
- (ii) What can be said to be the ultimate goal of intelligent agents? (1mark)
- (c) (i) Explain the concept "State space search". ✓ (1mark)
- (ii) Highlight three (3) Artificial Intelligence applications where "state space search" used? (3marks)
- (d) Define and give at least an example of each of the following key components and principles of state space search. (5marks)
- (i) State Representation ✓ (ii) State Space (iii) Operators/Actions

2bii
2cii

- 3(a) State the condition under which uninformed search can be used. ✓ (1mark)
- (b) (i) State one major demerit of uninformed search. ✓ (2marks)
- (ii) What do you understand by the term "informed search". ✓ (2marks)
- (iii) Why can the A* informed search algorithms, be said to be more effective? (2marks) ✓

(c) Explain the following concepts:

- (i) Heuristic Function ✓ (ii) Cost Function ✓ (2marks)

(d) Construct and label properly a state diagram for an on-line order. ✗ (6marks)

4(a) (i) Explain the concept "Tree Search". (3marks)

(ii) Highlight three (3) modern-day applications of tree search. (3marks)

(b)(i) State two (2) characteristics of the two (2) basics of a Tree Search. (2marks) ✓

(c) Describe the following Tree Search algorithms: (2marks) ✓

- (i) Depth-First Search (DFS) (ii) Breadth-First Search (BFS)

(iii) Highlight two (2) major application areas of the tree search. (2marks)

(d) (i) Identify and describe two (2) major challenges of the "Tree Search". (2marks)

(ii) What is "Knowledge Representation"? ✓ (1mark)

5(a) Explain the term "Backtracking". (2marks)

(b) (i) Highlight three (3) trending Artificial Intelligence Scheduling applications. (3marks)

(ii) Name the process by which a node is eliminated when it fails to meet a scheduling requirement? (1mark)

(c) (i) Describe the "Brute-Force" Approach. (2marks)

(ii) What is the difference between the "Brute-Force" Approach and the concept of "Dynamic Programming"? (2marks)

(d) Draw the state space tree for how the following three (3) events can be scheduled by a scheduling application making use the brute-force approach. (5marks)

- Reply my e-mails (RME)
- Go for shopping (GFS)
- Set-up a conference call (SCC)

6(a) Briefly describe the following key concepts and approaches in knowledge representation:

(i) Rule-Based Systems

(2marks)

(ii) Predicate Logic

(2marks)

(iii) Bayesian Networks

(2marks)

(b) (i) What factor determines the selection of an appropriate knowledge representation method for use?

(1mark)

(ii) State the goal of an Expert System. ✓

(2marks)

(c) Highlight two (2) programming languages used for Artificial Intelligence. (2marks) ✓

(d) Describe the following key concepts of Natural Language Processing.

(i) Tokenization ✓

(2marks)

(ii) Syntax and parsing

(2marks)



CALEB UNIVERSITY. LAGOS, NIGERIA.

Department of Computer Science
CSC 433: Human Computer Interactions
2023/2024 First Semester Examination

Time: 3hrs
3 Units
Marks: 60

INSTRUCTION: ANSWER ANY FOUR QUESTIONS.

QUESTION ONE

- i. Explain the consequences of poor interface design in Human-computer interaction (HCI) especially in respect to Computer Science. (5 marks).
- ii. HCI is a multidisciplinary subject covering computer science, behavioral sciences, cognitive science, ergonomics and human factors, psychology, Information security, Speech language pathology, Ethnography, Engineering and design principles. Write a short note on how HCI interfaces in any of these subjects, e.g HCI and Computer Science or HCI and Behavioral Sciences, etc. (5 marks).
- iii. Justify the reason for designing human-computer interfaces. (5 marks).

QUESTION TWO

- i. Explain what is meant by "innovative interfaces in everyday life". (5 marks).
- ii. Describe the roles of ergonomics in HCI. (5 marks).
- iii. Discuss data display guidelines. (5 marks).

QUESTION THREE

- i. Explain the four graph types that are available in most commercially available graphics packages. (4 marks).
- ii. Describe four of the Alternative Data Entry Techniques you are familiar with. (5 marks).
- iii. Data entry is notoriously error prone; hence, Data validation attempts to check that all mandatory fields are filled in, and that the data entered is correct, or at least reasonable. Explain four of the commonly used methods for data validation. (6 marks).

QUESTION FOUR

- i. Explain the steps involved in screen design. (5 marks)
- ii. One of the most important effects which has to be achieved when presenting information is to draw users' attention to important items. Discuss the techniques or visual attributes that can be used to highlight Data. (4 marks).
- iii. Explain human vision as it relates to computing. (3 marks).
- iv. Explain human learning and memory. (3 marks).

QUESTION FIVE

- i. Sitting for too long is associated with a higher risk of being diagnosed with, or dying of, breast, colon, colorectal, endometrial and ovarian cancers. Explain a way out of this in relation to Human Computer Interactions. (7 marks)
- ii. How does posture impact productivity? (4 marks).
- iii. What is Command line interface? (4 marks).

QUESTION SIX

- i. What role does user feedback plays in iterative design process? (5 marks)
- ii. How does the choice of input method affect user interaction and experience? (5 marks)
- iii. How can user experience be improved through interface design? (5 marks)



COLLEGE OF PURE AND APPLIED SCIENCES (COPAS)
DEPARTMENT OF COMPUTER SCIENCE

SESSION: 2023/2024

COURSE CODE: CSC 413

COURSE TITLE: COMPUTER GRAPHICS AND VISUALIZATION

INSTRUCTION: ATTEMPT FOUR QUESTIONS IN ALL

SEMESTER: FIRST
CREDIT UNIT: 3

TIME: 3 HOURS

Question One

- Computer graphics and visualization is an art of drawing pictures on computer screens with the help of programming. Describe what you understand by the term cathode ray tube (CRT) and Highlight the four (4) operation of cathode ray tube (5 marks) ✓
- By analogy ways through which object can be display on the screen used for raster graphics, differentiate between Raster Scan and Random Scan based on the following factors: (i) Electron Beam (ii) Resolution (iii) Picture Definition (iv) Realistic Display (v) Draw an image (5 marks) ✓
- Most of the presentation graphics has numerous applications page layouts to help in the creation of slides, exhaustively describe ten (10) out of computer graphics application (5 marks)

Question Two

- Concisely define what you understand by the terms Digital Differential Analyzer (DDA) algorithm. Inclines and clarify two (2) popular algorithms for generating a circle (3 marks)
- Deliberately describe the advantages and disadvantages of Digital Differential Analyzer (DDA) algorithm. (5 marks)
- Draw and display bresenham graph from any position (2, 3) (X_k, Y_k) you need to choose between (3, 3) ($X_k + 1, Y_k$) and (3, 4) ($X_k + 1, Y_k + 1$). You would like the point that is closer to the original line. (5 marks)
 $D \rightarrow \text{circle}$
 $\rightarrow \text{circle}$
- Identify the number of steps to put pixel. If $dx > dy$, then you need more steps in x coordinate; otherwise in y coordinate. (2 marks) ✓

Question Three

- Based on your understanding, explain in details what you understand by the terms clipping in computer graphics, list and illuminate two (2) popular algorithms for line clipping. (3 marks)
- Draw and clarify in details point clipping rectangle window, indicates X and Y-coordinate lies in between W_{x1} and W_{y1} . (5 marks)
- Contribute a brief explanation by the terms bitmap graphics, enumerate the main disadvantages of bitmap graphics (3 marks)

- d. Various techniques are used to provide text clipping in a computer graphics, identify three (3) methods for text clipping with the window boundary string before and after (4 marks)

Question Four

- a. Transformations are a fundamental part of computer graphics used to position objects, to shape objects, to change viewing positions. Enumerate and describe two (2) main types of transformation that one can perform in 2 dimensions with the matrix / vector representation of translations with the equation formula (8 marks)

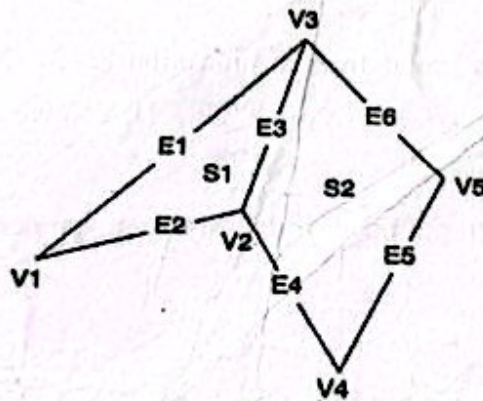
- b. Incline and clarify two (2) major types of parallel projection, display it and sub types (7 marks)

Question Five

- a. Succinctly describe the terms perspective projection in details, and identify three (3) types of perspective projection with the aid of flow diagram projection plane. (4 marks)
- b. Itemize four (4) properties of bazier curve (4 marks)
- c. Enumerate five (5) properties of b-spline curve (5 marks)
- d. The Bezier curve can be represented mathematically. Express the meaning of the following mathematical sentences: (i) $f(x, y) = 0$ (ii) $x^2 + y^2 - R^2 = 0$ (2 marks)

Question Six

- a. Highlight and describe two (2) 3D object representation. Study this figure below critically and provide the following information table: (i) Vertex table (ii) Vertex table (iii) Polygon surface table (8 marks)



- b. Identify and explain four (4) possible relationships that a surface can have with a specified area boundary. (4 marks)
- c. Explain the following types of computer graphics related to methods in surface detection: (i) Scan Line Method (ii) Area-Subdivision Method (3 marks)



CALEB UNIVERSITY
COLLEGE OF PURE AND APPLIED SCIENCES (COPAS)
DEPARTMENT OF COMPUTER SCIENCE
2023/2024 Session First Semester Examinations

CSC 411: Organization of Computer Languages
Instruction: Answer any four (4) questions

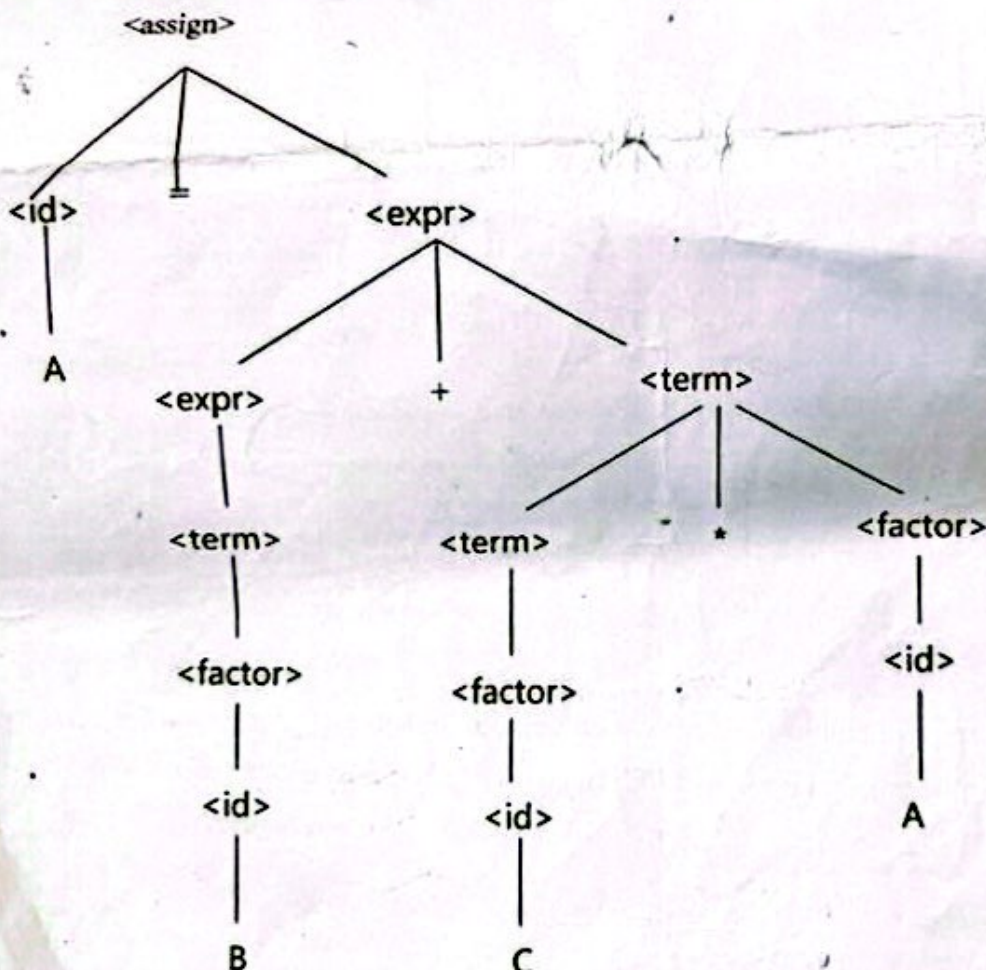
Time Allowed: 2 hours

Question 1

(a) Write-out the sentence for the Tree β . (2marks) ✓

(b) Write-out grammar that was used on the sentence referred to in (a) to derive the parse Tree β . (3marks) ✓

TREE β



- (c) Give a simple but concise definition of an ambiguous grammar. ✓ (3marks)
- (d) State the reason why the Grammar below can be considered an ambiguous grammar? ✓ (3marks)

$\langle \text{assign} \rangle \Rightarrow \langle \text{id} \rangle = \langle \text{expr} \rangle$

$\Rightarrow \langle \text{id} \rangle = A \mid B \mid C$

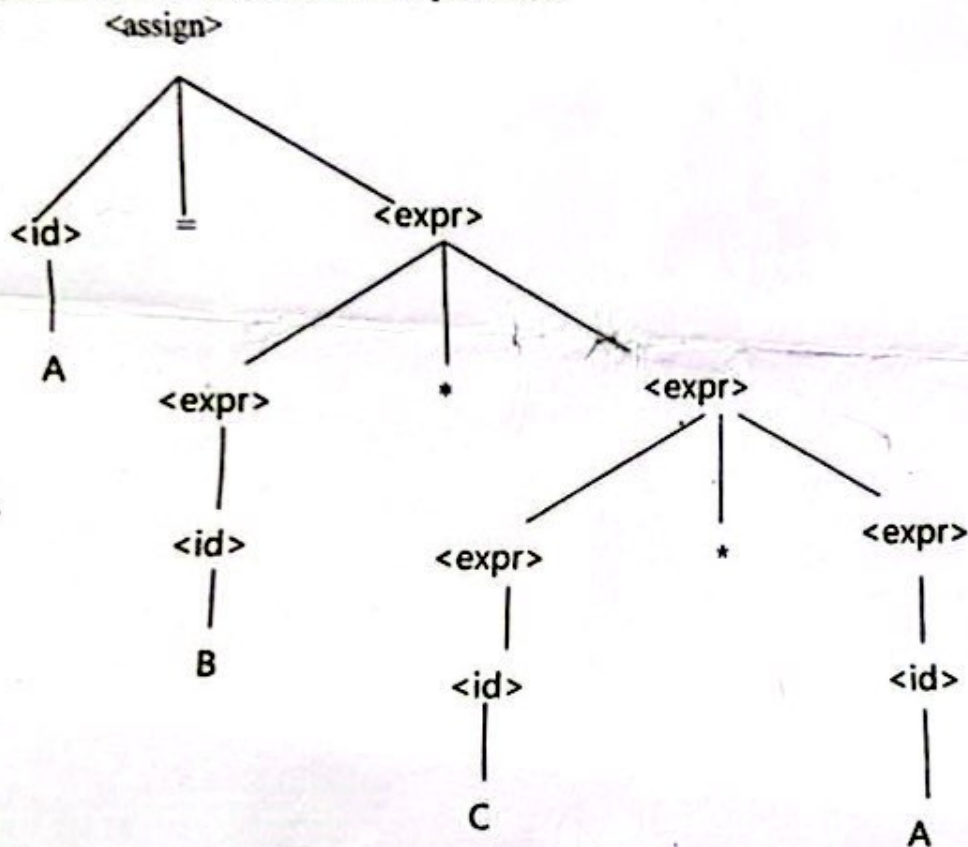
$\Rightarrow \langle \text{expr} \rangle = \langle \text{expr} \rangle + \langle \text{expr} \rangle$

$\mid \langle \text{expr} \rangle * \langle \text{expr} \rangle$

$\mid \langle \text{expr} \rangle$

$\mid \langle \text{id} \rangle$

- (e) If the Parse tree below represents one of two distinct parse trees for an ambiguous grammar, draw the other distinct parse tree. (4marks) ✓



Question 2

- (a) Highlight the type of problem associated with language structures. (1mark)
- (b) Describe how the problem associated with language structure affect code generation in compilers. (2marks)

(c) Some parsing algorithms (parser) can be based on ambiguous grammars. Describe how such parsers deal with the problem of ambiguous constructs? (3marks)

(d) Identify and state the semantic problem in the expression below: (4marks)

$$\Theta + \gamma * \alpha$$

(e) Provide a detailed explanation of how a parser would solve the semantic problem stated in question 2(d) above. (5marks)

Question 3

(a) Describe process of sentence generation, indicating the nexus between this process and the concept of "derivation". (2marks)

(b) In the following program statement: $\text{index} = 2 * \text{count} + 17$, identify all: (7marks)

(i) Lexemes ✓

(ii) Tokens ✓

(c) State the main function of the "Backus-Naur Form" notation. (1mark)

(d) Describe how a parse trees is related to formal grammar. (2marks)

(e) Given the following rules below, provide a clear and concise step-wise derivation of the natural number 899. (3marks)

$\langle \text{nat} \rangle \Rightarrow \langle \text{digit} \rangle / \langle \text{nonzero} \rangle \langle \text{digits} \rangle$

$\langle \text{digits} \rangle \Rightarrow \langle \text{digit} \rangle / \langle \text{digit} \rangle \langle \text{digits} \rangle$

$\langle \text{digit} \rangle \Rightarrow 0 / \langle \text{nonzero} \rangle$

$\langle \text{nonzero} \rangle \Rightarrow 1|2|3|4|5|6|7|8|9$

Question 4

(a) (i) Use the Grammar below to process the expression $c = a + 10$; and (1mark) ✓

(ii) Draw the corresponding parse tree. (5marks)

Grammar:

$\langle \text{assignment-stmt} \rangle ::= \langle \text{identifier} \rangle = \langle \text{expr} \rangle;$

$\langle \text{expr} \rangle ::= \langle \text{expr} \rangle + \langle \text{expr} \rangle \mid \langle \text{expr} \rangle - \langle \text{expr} \rangle \mid (\langle \text{expr} \rangle) \mid \langle \text{identifier} \rangle \mid \langle \text{integer} \rangle$

$\langle \text{identifier} \rangle = a \mid c$

(b) Define the following terms:

(i) Syntax (1mark) ✓

(ii) Grammar (1mark) ✓

(iii) Semantics (1mark) ✓

- (c) State four (4) attributes of good programming language (2marks) ✓
 (d) Explain the difference between terminals and non-terminals in BNF. (1marks) ✓
 (e) Highlight three (3) elements of a lexical structure. (3marks) ✓

Question 5

- (a) Give a brief explanation each of the following: (2marks)
 (i) The root of a parse tree. ✓
 (ii) The leaves of a parse tree. ✓
 (b) If the Grammar below is used to process the expression $E = mc^2$, draw the parse tree that will be obtained. (6marks) ✓

Grammar:

$\langle \text{assignment-stmt} \rangle ::= \langle \text{identifier} \rangle = \langle \text{expr} \rangle;$

$\langle \text{expr} \rangle ::= \langle \text{expr} \rangle + \langle \text{expr} \rangle \mid \langle \text{expr} \rangle - \langle \text{expr} \rangle \mid \langle \text{expr} \rangle * \langle \text{expr} \rangle \mid \langle \text{expr} \rangle / \langle \text{expr} \rangle \mid$

$(\langle \text{expr} \rangle) \mid \langle \text{identifier} \rangle \mid \langle \text{integer} \rangle$

$\langle \text{identifier} \rangle = E \mid m \mid C$

- (c) State the principle of operator precedence. (1mark) ✓
 (d) (i) Write-out and name the specific operators that determine left associativity. (4marks)
 (ii) Write-out and name the specific operators that determine right associativity. (2marks)

Question 6

- (a) Demonstrate that the following grammar is ambiguous (Capitals are non-terminals, lowercase are terminals): (7marks)

$S \rightarrow A a B \mid B a A$

$A \rightarrow b \mid c$

$B \rightarrow a \mid b$

- (b) Write an unambiguous grammar generating the set of all strings over the alphabet $\{0, 1, +, -\}$, where $+$ and $-$ are infix operators which both associate to the left and such that $+$ binds more tightly than $-$. (2marks)
 (c) For each of the following descriptions, give a regular expression over the alphabet $\{a, b, c\}$, and a regular grammar that generates the language described. a.
 The set of all strings over $\{a, b, c\}$, where each string has at most one a.
 (i) The set of all strings over $\{a, b, c\}$, where each string has at most one a (2marks)
 (ii) The set of all strings over $\{a, b, c\}$, where, in each string, every b is immediately followed by at least one c. (2marks)
 (d) Write an expression over the alphabet $\{a, b, c\}$ for all set of all strings over $\{a, b, c\}$, where every string has length a multiple of four. (2marks)